

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity sixteen_to_fourEncoder is
5      port(input3, input2, input1, input0:    in std_logic_vector(3 downto 0);
6          s:                                  in std_logic_vector(1 downto 0);
7          en:                                 in std_logic;
8          output16_4En:                       out std_logic_vector(3 downto 0));
9  end;
10
11 architecture synth of sixteen_to_fourEncoder is
12 begin
13     process (s, en) begin
14         if (s(1) = '1' and s(0) = '1' and en = '1') then
15             output16_4En(3 downto 0) <= input3(3 downto 0);
16         elsif (s(1) = '1' and s(0) = '0' and en = '1') then
17             output16_4En(3 downto 0) <= input2(3 downto 0);
18         elsif (s(1) = '0' and s(0) = '1' and en = '1') then
19             output16_4En(3 downto 0) <= input1(3 downto 0);
20         elsif (s(1) = '0' and s(0) = '0' and en = '1') then
21             output16_4En(3 downto 0) <= input0(3 downto 0);
22         else
23             output16_4En(3 downto 0) <= input0(3 downto 0);
24         end if;
25     end process;
26 end;
```